

VisualSpatial distance

- The VisualSpatial distance is a custom metric designed to scale the spatial distance based on the visual similarity of two objects. This means that the SpatialDistance is still dominant but slightly affected based on how similar two objects are.
- If two objects are visually identical, we scale down the spatial distance up to a minimum (minimum spatial distance factor, currently 50%)
- If two objects are similar we don't apply any scaling (two objects are similar if their visual similarity is equal to the critical visual distance, currently 0.25)
- To improve the reliability of the visual features we are using the visual scaling factor at a 50% rate. (Visual Scaling contribution factor)

$(0.5 + 2.0 * (0.5 + 0.5 * \text{visualScalingFactor}) * \text{visualDistance}) * \text{spatialDistance}$

Minimum spatial distance factor = 0.5 (i.e. 50%)

If two objects are visually identical, we still consider their spatial distance during association but scaled to 50%

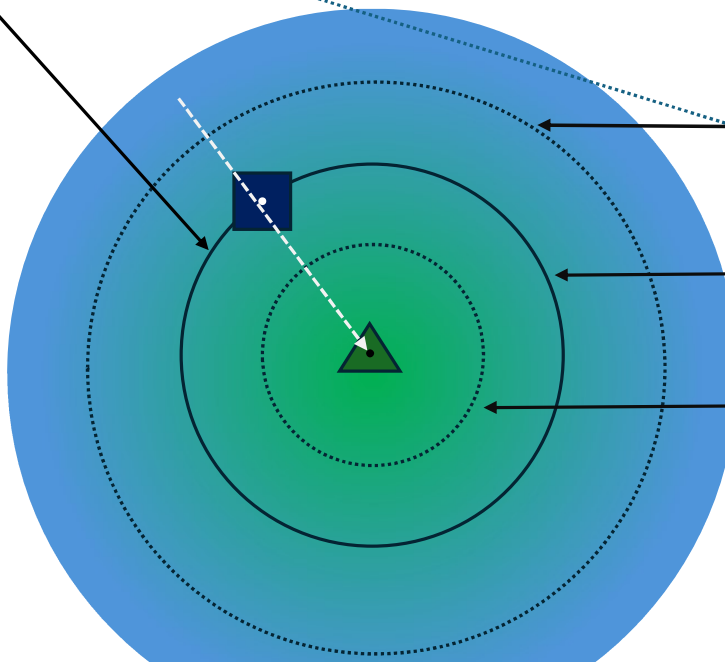
Visual Scaling contribution factor = 0.5 (i.e. 50%)

This controls how much does the visual scaling factor affect the visual features

Critical Visual Distance value = 0.25

At this value the visual distance has practically no effect over the spatial distance

$$\frac{1 - \text{Minimum spatial distance factor}}{\text{Critical Visual Distance value}} = \frac{0.5}{0.25} = 2$$



visual_distance = 0.50
Spatial distance scaling = 1.5x

visual_distance = 0.25
Spatial distance is scaling = 1x

visual_distance = 0.00
Spatial distance scaling = 0.5x

```
(MS + (1 - MS)/(CVD)*((1 - VS) + VS * visualScalingFactor) * visualDistance) * spatialDistance
```

MS - Minimum spatial distance factor = 0.5 (i.e. 50%)

VS - Visual Scaling contribution factor = 0.5 (i.e. 50%)

CVD - Critical Visual Distance value = 0.25